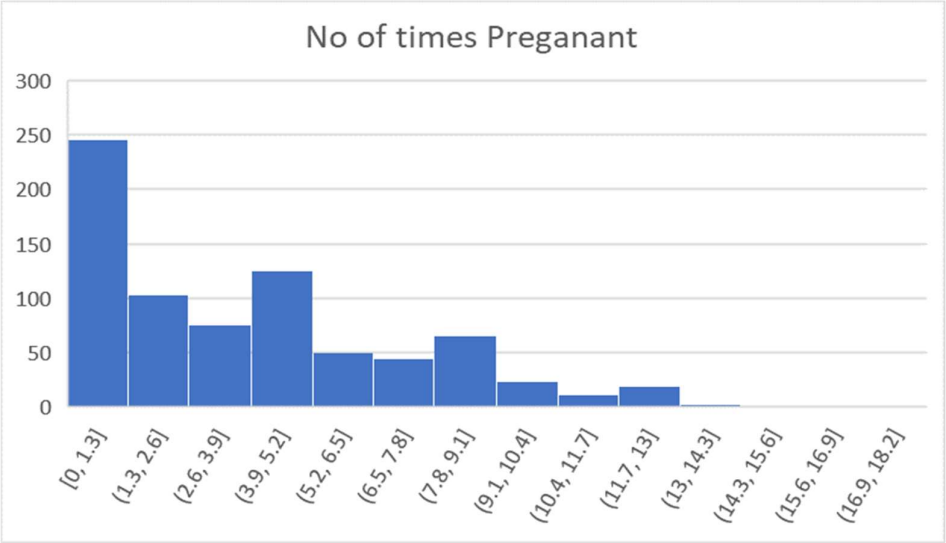
Name – Rahul Bhujade

Model for Diabetes dataset

Since this is the classification problem with only 2 responses as a target. We can build a binary logistic regression model.

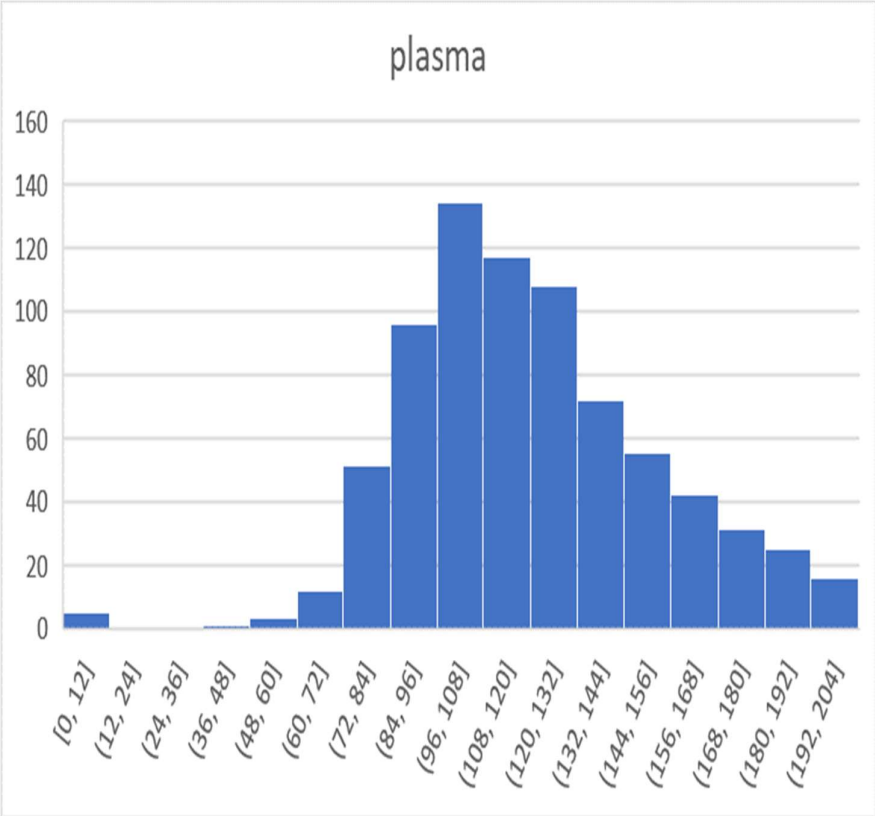
Data description:

1. No of times Pregnant



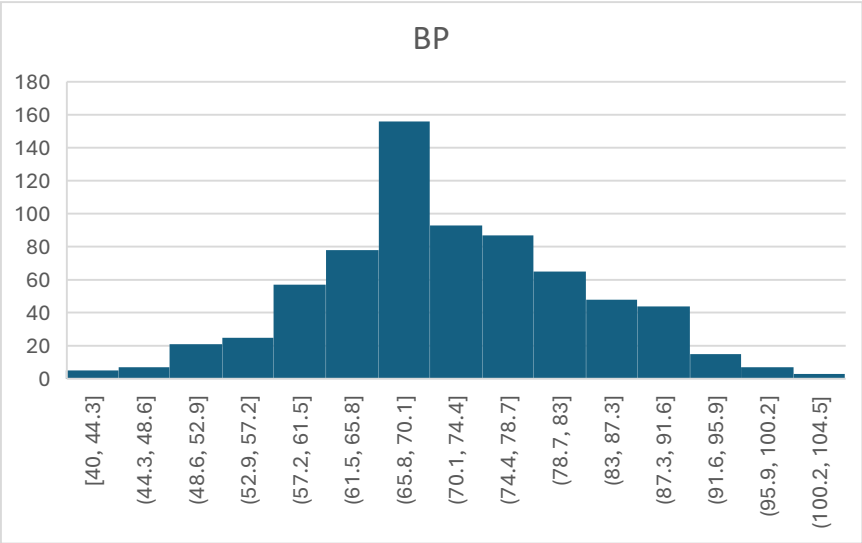
times Preganant	
Mean	3.845052083
Standard E	0.121589175
Median	3
Mode	1
Standard D	3.369578063
Sample Va	11.35405632
Kurtosis	0.159219778
Skewness	0.901673979
Range	17
Minimum	0
Maximum	17
Sum	2953
Count	768

2. Plasma



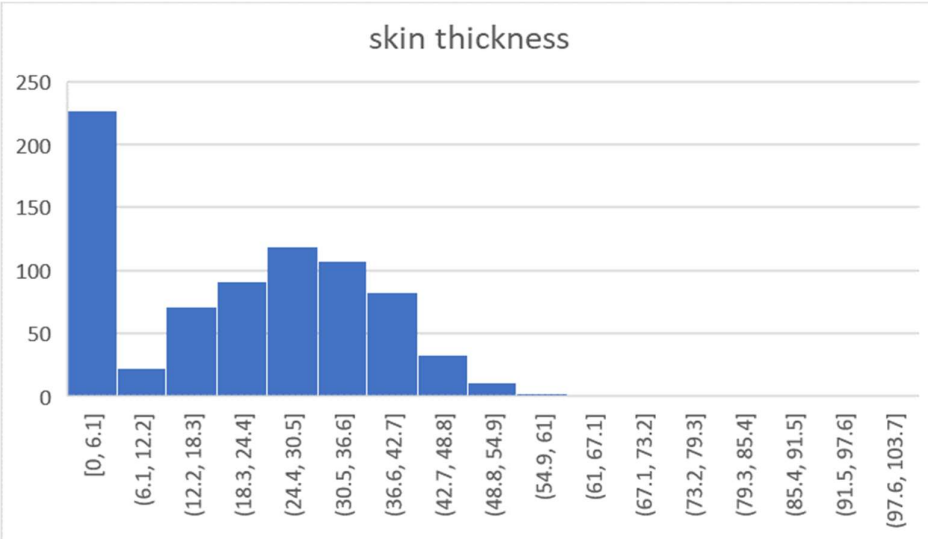
Plasma	
Mean	120.8945313
Standard E	1.153712483
Median	117
Mode	100
Standard D	31.9726182
Sample Va	1022.248314
Kurtosis	0.64077982
Skewness	0.173753502
Range	199
Minimum	0
Maximum	199
Sum	92847
Count	768

3. BP



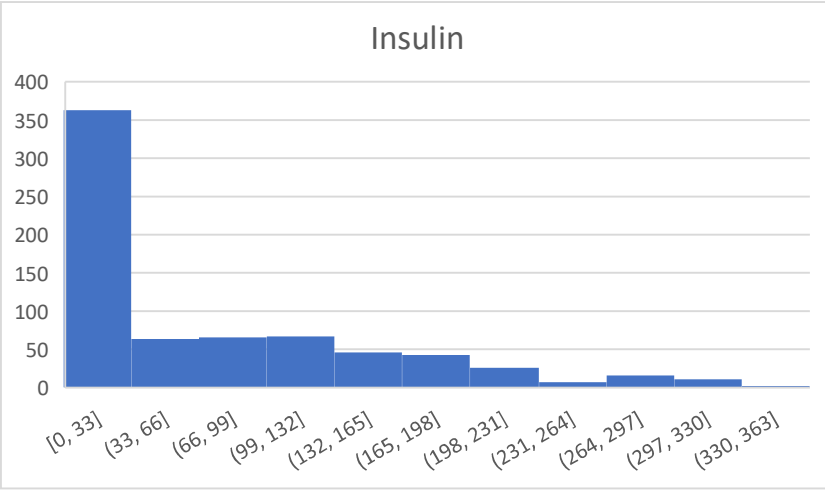
BP	*SI*
Mean	69.10546875
Standard Error	0.69844253
Median	72
Mode	70
Standard Deviation	19.35580717
Sample Variance	374.6472712
Kurtosis	5.18015656
Skewness	-1.843607983
Range	122
Minimum	0
Maximum	122
Sum	53073
Count	768

4. Skin Thickness



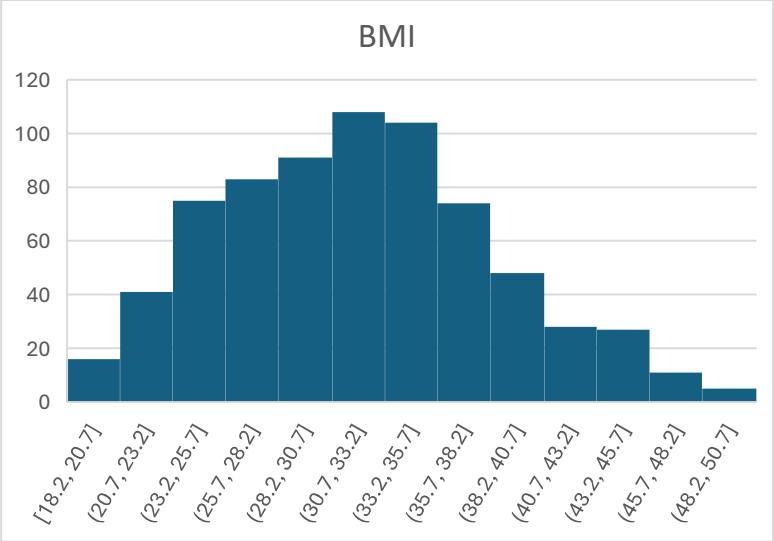
kinThickness	
Mean	20.53645833
Standard E	0.575626069
Median	23
Mode	0
Standard D	15.95221757
Sample Va	254.4732453
Kurtosis	-0.520071866
Skewness	0.109372496
Range	99
Minimum	0
Maximum	99
Sum	15772
Count	768

1. Insulin



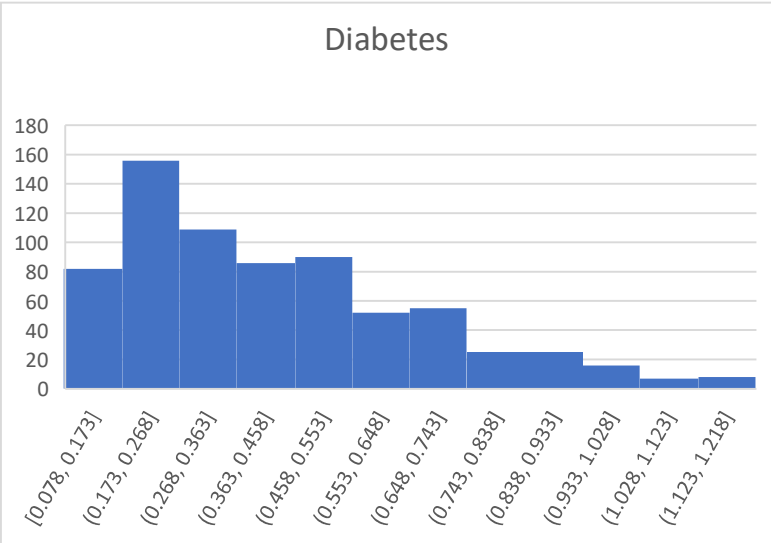
Insulin	
Mean	79.79948
Standard E	4.15851
Median	30.5
Mode	0
Standard D	115.244
Sample Va	13281.18
Kurtosis	7.21426
Skewness	2.272251
Range	846
Minimum	0
Maximum	846
Sum	61286
Count	768

2. BMI



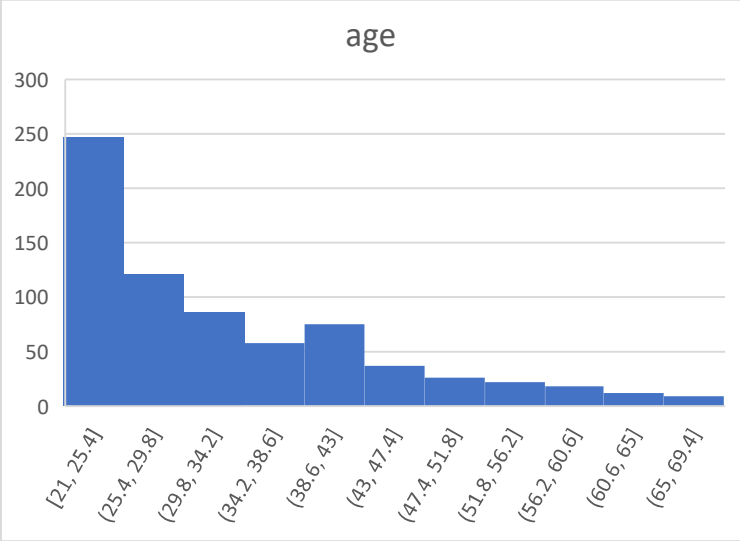
BMI	
Mean	31.99257813
Standard E	0.28449513
Median	32
Mode	32
Standard D	7.88416032
Sample Va	62.15998396
Kurtosis	3.290442901
Skewness	-0.428981588
Range	67.1
Minimum	0
Maximum	67.1
Sum	24570.3
Count	768

3. Diabetes



Diabetes	
Mean	0.471876302
Standard E	0.011955791
Median	0.3725
Mode	0.254
Standard D	0.331328595
Sample Va	0.109778638
Kurtosis	5.594953528
Skewness	1.919911066
Range	2.342
Minimum	0.078
Maximum	2.42
Sum	362.401
Count	768

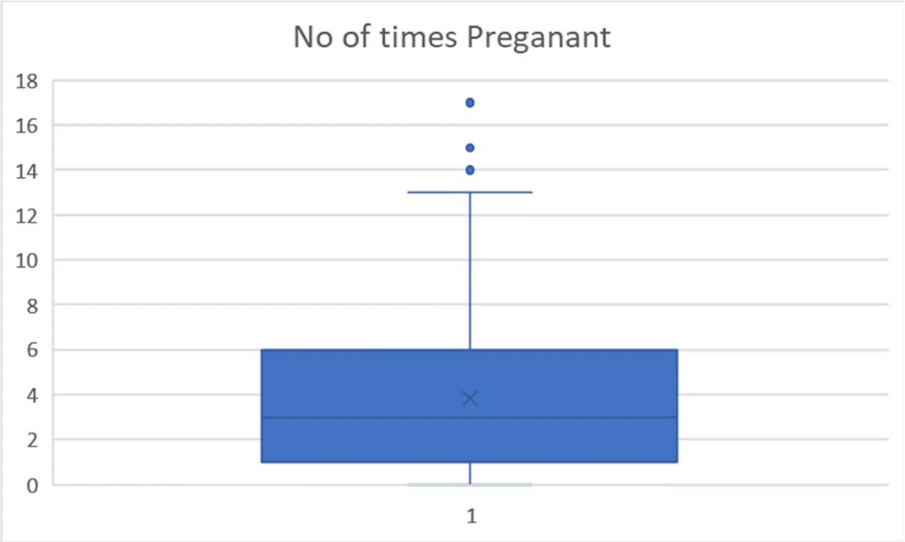
4. Age

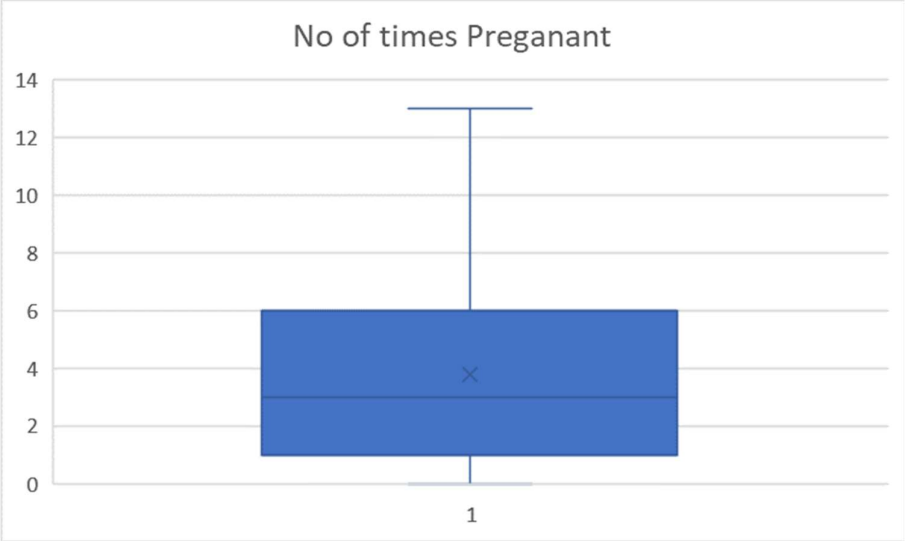


Age	
Mean	33.24088542
Standard E	0.424360803
Median	29
Mode	22
Standard D	11.76023154
Sample Va	138.3030459
Kurtosis	0.643158889
Skewness	1.129596701
Range	60
Minimum	21
Maximum	81
Sum	25529
Count	768

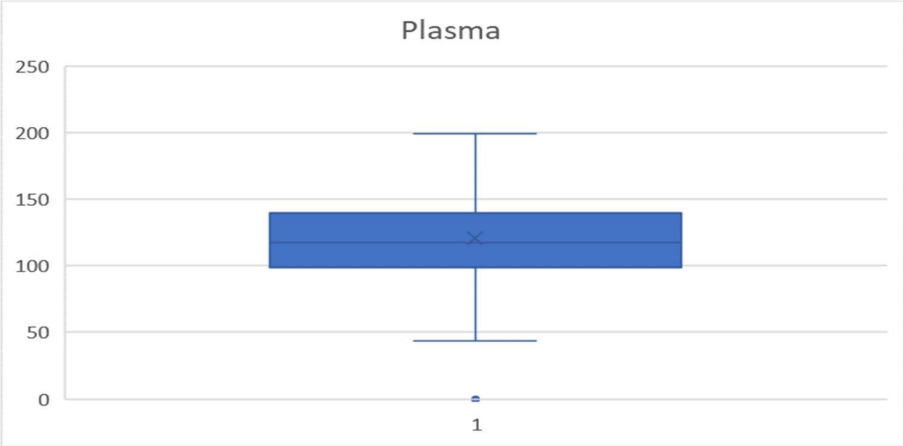
Boxplot and outlier detection:

1. **Pregnant**

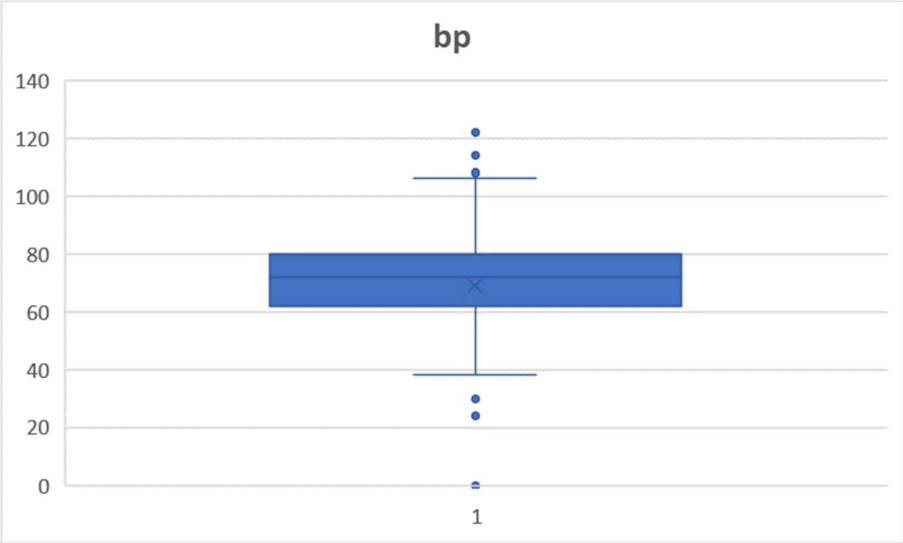




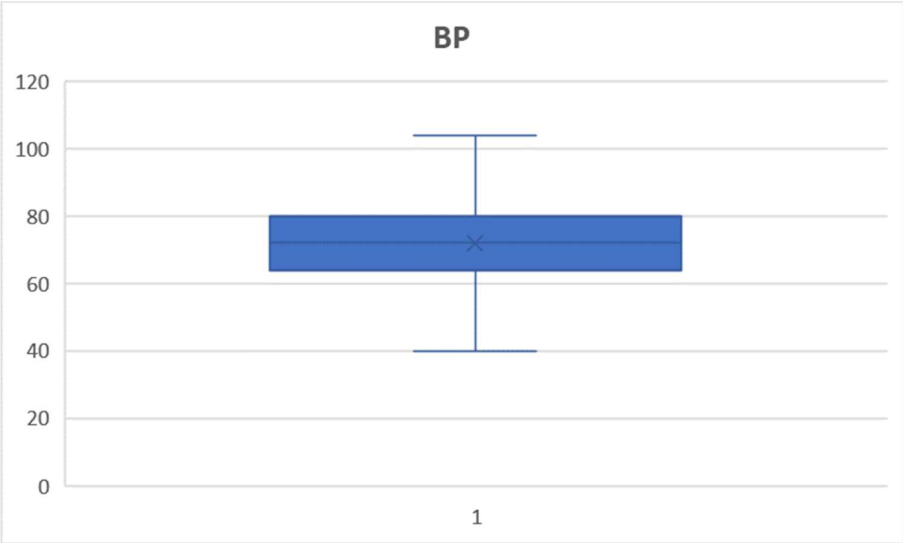
2.plasma

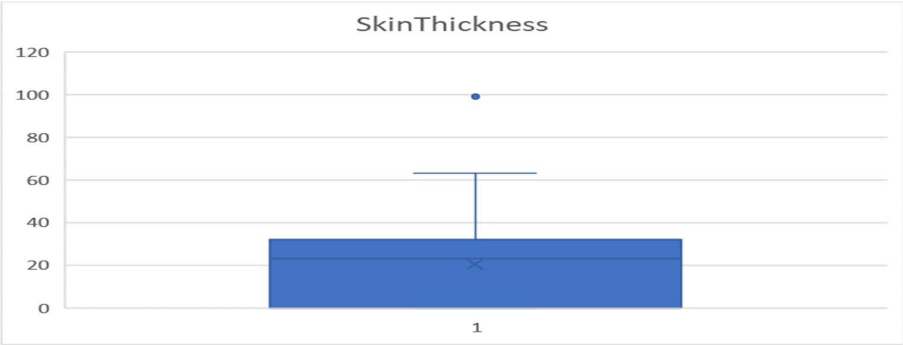


3.BP

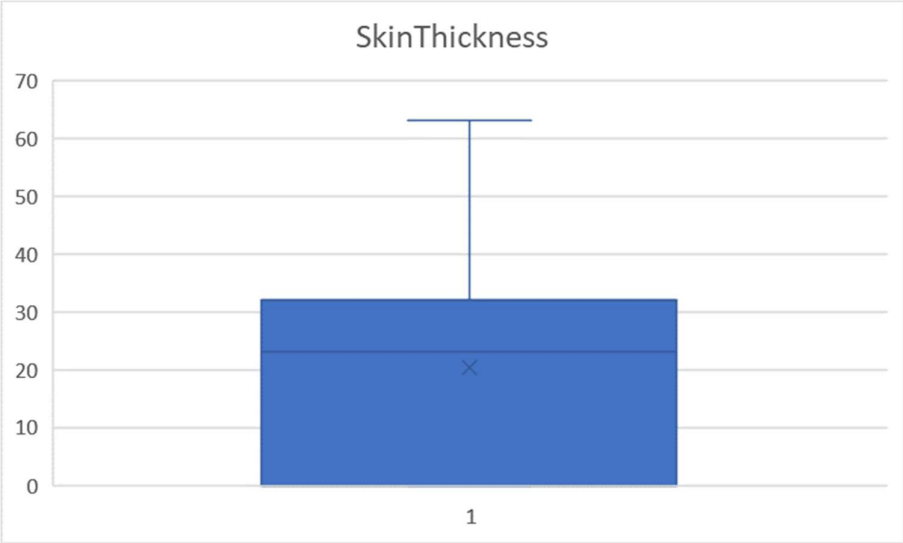


After removing outliers by replacing it with mean i.e. 68

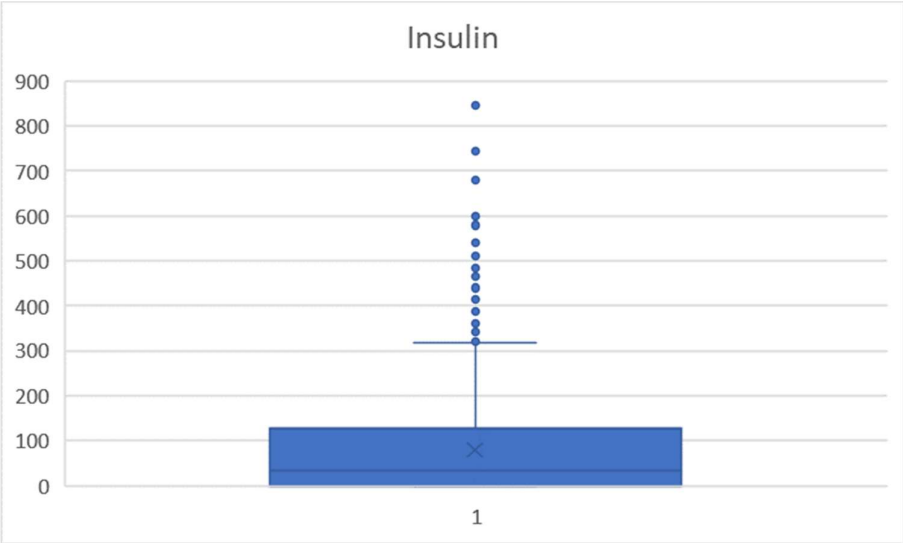




Removing only one outlier

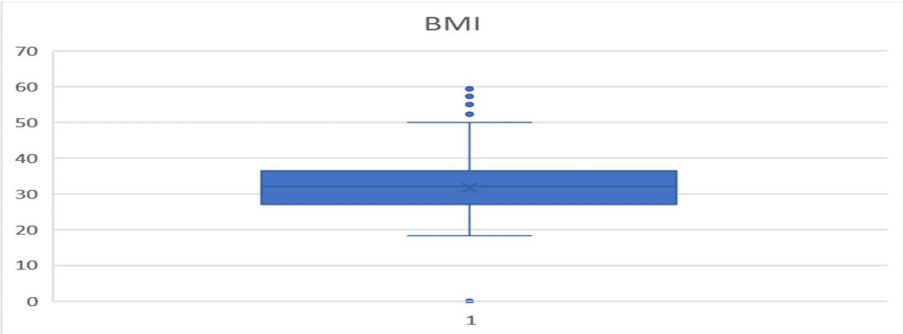


6. Insulin

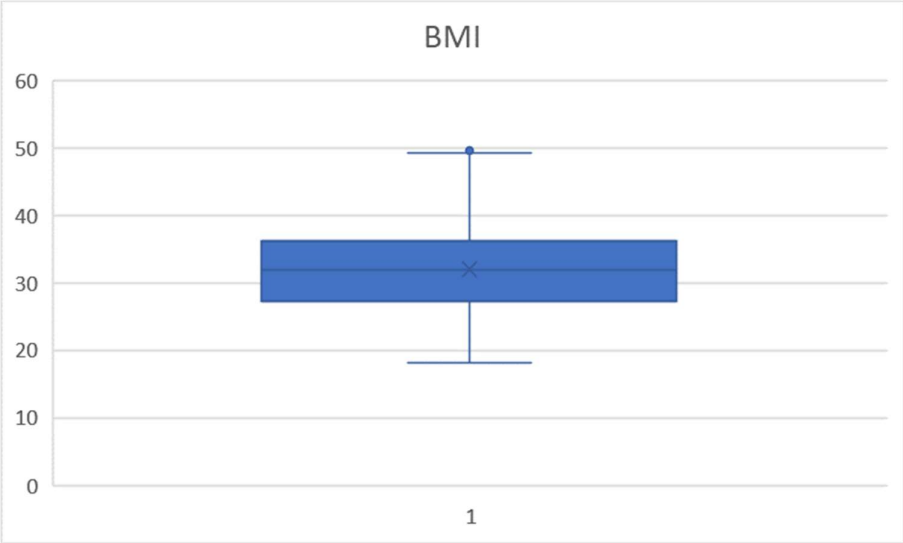


There were about 26 outliers removed all of them

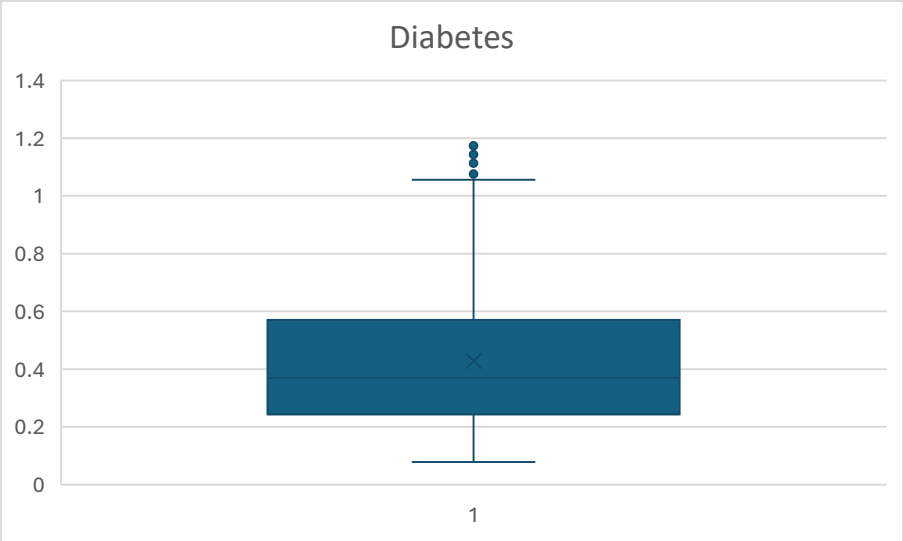
7.BMI



Removed 15 values

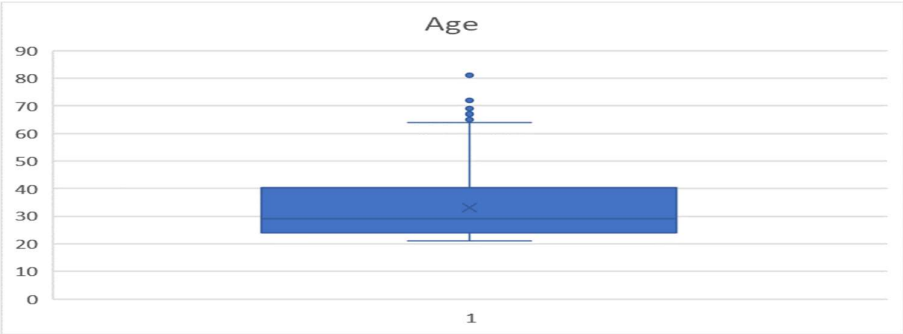


8. Diabetes



Replacing outlier values with mean value

9. Age



Removed only extreme outliers in age

Correlation matrix

	No of times Preganant	*Plasma*	*BP*	*SkinThickness*	*Insulin*	*BMI*	*Diabetes*	*Age*	*Class*
No of time	1								
Plasma	0.135994942	1							
BP	0.196586349	0.238970758	1						
SkinThickn	-0.075673248	-0.018470702	0.037881865	1					
Insulin	-0.083470604	0.245583998	-0.040980323	0.492718833	1				
BMI	0.02181538	0.205117949	0.271679212	0.378018997	0.176217	1			
Diabetes	-0.004333354	0.055376184	0.043942528	0.15981619	0.20879	0.145502429	1		
Age	0.560668905	0.267432266	0.342938372	-0.146458152	-0.07481	0.057055706	0.047036	1	
Class	0.205266759	0.462539136	0.170865134	0.039617894	0.090924	0.294263151	0.159025	0.234229199	1

Removing skin thickness attribute and no of times pregnant attributes

Model extraction:

Logistic Regression

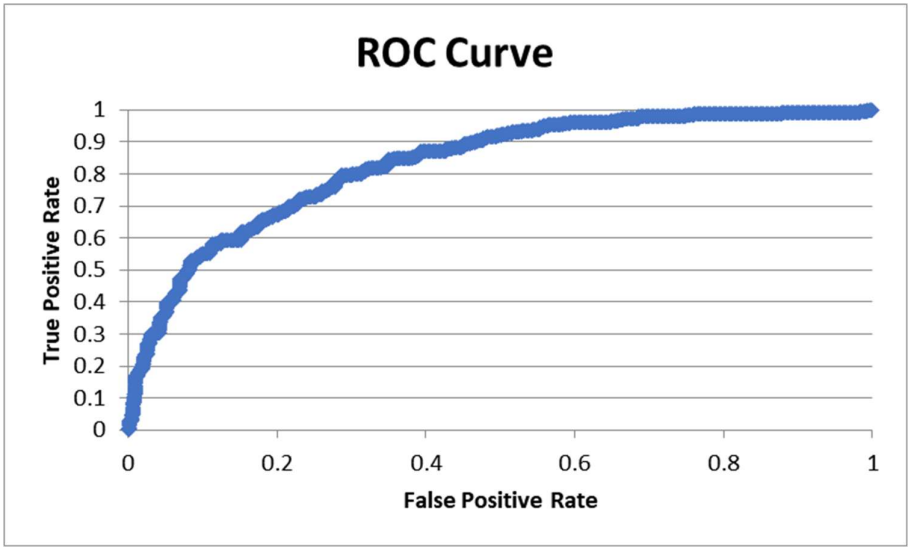
	# Iter	20			Alpha	0.05	
	coeff	*s.e.*	*Wald*	*p-value*	*exp(b)*	*lower*	*upper*
intercept	-9.08646	0.871788	108.6345	0	0.000113		
Plasma	0.036369	0.003844	89.52533	0	1.037038	1.029255	1.044881
BP	-0.00761	0.00959	0.62906	0.4277	0.992423	0.973943	1.011253
Insulin	-0.00182	0.001157	2.483258	0.115064	0.998178	0.995916	1.000445
BMI	0.092917	0.016588	31.37769	2.12E-08	1.097371	1.062268	1.133634
Diabetes	1.343862	0.392895	11.69922	0.000625	3.833822	1.775004	8.280653
Age	0.02778	0.008915	9.711005	0.001832	1.02817	1.010361	1.046293

BP and insulin come as a insignificant variables

Model evaluation and diagnostics:

Classification Table

Classification Table				
	Obs Suc	Obs Fail	Total	
Pred Suc	134	52	186	
Pred Fail	102	423	525	
Total	236	475	711	
Accuracy	0.567797	0.890526	0.783404	
Cutoff	0.5			
AUC	0.829411			



LL statistics

LL	-337.768
LL0	-451.865
Chi-sq	228.194
df	6
p-value	0
R-sq (L)	0.252502
R-sq (CS)	0.274539
R-sq (N)	0.381586
AIC	689.5365
BIC	721.5032

Overall accuracy of the model come to be 0.78 with AUC value of 0.82 at cut-off of 0.5. Therefore, we can say that our model is moderately accurate to predict the class of the sample value.